

SUPPLEMENTARY MATERIALS

Smart Indoor Lighting for Sustainable Buildings: A Systematic Bibliometric Review of Human-Centric Control, IoT Platforms, and Automation-Related Optimization

S1. Complete Database-Specific Search Strings per Block

S1.1 Scopus Search Strings

Block	Search String
Block A	TITLE-ABS-KEY(("human-centric lighting" OR "circadian lighting" OR "adaptive lighting" OR "visual comfort"))
Block B	TITLE-ABS-KEY(("energy efficient lighting" OR "energy saving lighting" OR "lighting optimization"))
Block C (Strict)	TITLE-ABS-KEY(("hyperheuristic" OR "hyper-heuristic" OR metaheuristic*) AND (lighting OR "illumination control"))
Block C (Expanded)	TITLE-ABS-KEY(("hyperheuristic" OR "hyper-heuristic" OR metaheuristic* OR "algorithm selection" OR "high-level heuristic" OR "heuristic selection") AND (lighting OR "illumination control" OR "smart lighting"))
Block D	TITLE-ABS-KEY(("Internet of Things" OR IoT OR Zigbee OR DALI OR MQTT OR KNX) AND ("smart lighting" OR "indoor lighting"))
Block E (Strict)	TITLE-ABS-KEY(("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation") AND (lighting OR "indoor lighting"))

Block E (Expanded)	TITLE-ABS-KEY(("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation" OR "workflow automation" OR "trigger-action" OR "rule engine") AND (lighting OR "indoor lighting" OR "smart lighting"))
Block E (Middleware- enriched)	TITLE-ABS-KEY(("IFTTT" OR "if this then that" OR "event-condition-action" OR "Node-RED" OR "Home Assistant" OR "MQTT") AND (lighting OR "indoor lighting" OR "smart lighting"))

S1.2 Web of Science Search Strings

Block	Search String
Block A	TS=("human-centric lighting" OR "circadian lighting" OR "adaptive lighting" OR "visual comfort")
Block B	TS=("energy efficient lighting" OR "energy saving lighting" OR "lighting optimization")
Block C (Strict)	TS=("hyperheuristic" OR "hyper-heuristic" OR metaheuristic*) AND TS=(lighting OR "illumination control")
Block C (Expanded)	TS=("hyperheuristic" OR "hyper-heuristic" OR metaheuristic* OR "algorithm selection" OR "high-level heuristic" OR "heuristic selection") AND TS=(lighting OR "illumination control" OR "smart lighting")
Block D	TS=("Internet of Things" OR IoT OR Zigbee OR DALI OR MQTT OR KNX) AND TS=("smart lighting" OR "indoor lighting")
Block E (Strict)	TS=("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation") AND TS=(lighting OR "indoor lighting")
Block E (Expanded)	TS=("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation" OR "workflow automation" OR "trigger-action" OR "rule engine") AND TS=(lighting OR "indoor lighting" OR "smart lighting")

Block E (Middleware- enriched)	TS=("IFTTT" OR "if this then that" OR "event-condition-action" OR "Node-RED" OR "Home Assistant" OR "MQTT") AND TS=(lighting OR "indoor lighting" OR "smart lighting")
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S1.3 IEEE Xplore Search Strings

Block	Search String
Block A	("human-centric lighting" OR "circadian lighting" OR "adaptive lighting" OR "visual comfort")
Block B	("energy efficient lighting" OR "energy saving lighting" OR "lighting optimization")
Block C (Strict)	("hyperheuristic" OR "hyper-heuristic" OR metaheuristic*) AND (lighting OR "illumination control")
Block C (Expanded)	("hyperheuristic" OR "hyper-heuristic" OR metaheuristic* OR "algorithm selection" OR "high-level heuristic" OR "heuristic selection") AND (lighting OR "illumination control" OR "smart lighting")
Block D	("Internet of Things" OR IoT OR Zigbee OR DALI OR MQTT OR KNX) AND ("smart lighting" OR "indoor lighting")
Block E (Strict)	("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation") AND (lighting OR "indoor lighting")
Block E (Expanded)	("IFTTT" OR "if this then that" OR "event-condition-action" OR "rule-based automation" OR "workflow automation" OR "trigger-action" OR "rule engine") AND (lighting OR "indoor lighting" OR "smart lighting")
Block E (Middleware- enriched)	("IFTTT" OR "if this then that" OR "event-condition-action" OR "Node-RED" OR "Home Assistant" OR "MQTT") AND (lighting OR "indoor lighting" OR "smart lighting")

S2. Stopword Lists, Stemming Rules, and Deduplication Protocols

S2.1 Stopword List (Excluded Terms)

The following stopwords were excluded from keyword analysis to reduce noise and eliminate non-substantive terms from bibliometric co-occurrence calculations:

a, about, above, across, after, afterwards, again, against, all, almost, alone, along, already, also, although, always, am, among, amongst, amount, an, and, another, any, anyhow, anyone, anything, anyway, anywhere, are, around, as, at, back, be, became, because, become, becomes, becoming, been, before, beforehand, behind, being, below, beside, besides, between, beyond, bill, both, bottom, but, by, call, can, cannot, cant, co, con, could, couldnt, cry, de, describe, detail, do, done, down, due, during, each, eg, eight, either, eleven, else, elsewhere, empty, etc, even, ever, every, everyone, everything, everywhere, except, few, fifteen, fifty, fill, find, fire, first, five, for, former, formerly, forty, found, four, from, front, full, further, get, give, go, had, has, hasnt, have, he, hence, her, here, hereafter, hereby, herein, hereupon, hers, herself, him, himself, his, how, however, hundred, ie, if, in, inc, indeed, interest, into, is, it, its, itself, keep, last, latter, latterly, least, less, ltd, made, many, may, me, meanwhile, might, mill, mine, more, moreover, most, mostly, move, much, must, my, myself, name, namely, neither, never, nevertheless, next, nine, no, nobody, none, noone, nor, not, nothing, now, nowhere, of, off, often, on, once, one, only, onto, or, other, others, otherwise, our, ours, ourselves, out, over, own, part, per, perhaps, please, put, rather, re, same, see, seem, seemed, seeming, seems, serious, several, she, should, show, side, since, sincere, six, sixty, so, some, somehow, someone, something, sometime, sometimes, somewhere, still, such, system, take, ten, than, that, the, their, them, themselves, then, thence,

there, thereafter, thereby,
therefore, therein, thereupon, these, they, thick, thin, third, this,
those, though, three,
through, throughout, thru, thus, to, together, too, top, toward,
towards, twelve, twenty, two,
un, under, until, up, upon, us, very, via, was, we, well, were, what,
whatever, when, whence,
whenever, where, whereafter, whereas, whereby, wherein, whereupon,
wherever, whether, which,
while, whither, who, whoever, whole, whom, whose, why, will, with,
within, without, would,
yet, you, your, yours, yourself, yourselves

S2.2 Stemming Rules

Term normalisation was applied using the **Porter Stemming algorithm**, with the following rule categories applied systematically:

Rule Category	Description	Example
Plural removal	Remove 's' from plural forms	"lightings" → "lighting"
Past tense removal	Remove 'ed' where appropriate	"optimized" → "optimize"
Present participle removal	Remove 'ing' where appropriate	"controlling" → "control"
Comparative removal	Remove 'er', 'est'	"brighter" → "bright"
Derivational suffix removal	Remove 'tion', 'ment', 'ness', etc.	"optimization" → "optimize"
Adverbial removal	Remove 'ly' where appropriate	"efficiently" → "efficient"

S2.3 Deduplication Protocol

Duplicate records were identified and removed using a **four-stage hierarchical protocol**:

Priority	Field	Method	Description
1	DOI	Exact match (case-insensitive)	Records sharing identical DOI are considered duplicates

2	Title + Year	Exact match after normalisation	Records with identical normalised titles and publication year are considered duplicates
3	Title + Author(s)	Matching first author + title similarity > 95%	Records sharing the same first author and highly similar titles (>95% similarity via Levenshtein distance) are considered duplicates
4	Multiple fields	Fuzzy matching on title, author, and source	Records with fuzzy matches across title, author list, and source are manually reviewed for duplication

Deduplication Workflow

1. Export records from Scopus, Web of Science, and IEEE Xplore in CSV/BibTeX format.
 2. Normalise all fields (trim whitespace, unify case, standardise punctuation).
 3. Extract DOI for each record; group records by DOI → remove exact duplicates.
 4. For remaining records, group by normalised Title + Year → remove duplicates.
 5. For remaining records, group by normalised Title + First Author surname → remove duplicates where similarity exceeds 95%.
 6. For remaining potential duplicates, apply fuzzy matching (Levenshtein distance < 5%) and manually review for duplication.
 7. **Total duplicate records identified and removed: 300.**
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S3. VOSviewer Parameter Files and Thesaurus

S3.1 VOSviewer Configuration Parameters

Software: VOSviewer version 1.6.20

Parameter	Block A	Block B	Block D
Analysis Type	Co-occurrence	Co-occurrence	Co-occurrence
Unit of Analysis	Keywords	Keywords	Keywords
Counting Method	Full counting	Full counting	Full counting

Minimum Occurrences	5	5	3*
Number of Keywords	238	156	89
Attraction	0.5	0.5	0.5
Repulsion	-1.0	-1.0	-1.0
Layout Algorithm	VOSlayout	VOSlayout	VOSlayout
Normalisation Method	Association strength	Association strength	Association strength
Clustering Resolution	1.00	1.00	1.00
Minimum Cluster Size	3	3	2

Note: The lower threshold for Block D (minimum occurrence = 3) was applied to preserve lower-frequency but technologically important platform-specific terms (e.g., Thread, Home Assistant, Node-RED) that are essential for capturing emerging IoT and middleware technologies.

S3.2 Thesaurus File (Keyword Normalisation)

File Format: thesaurus.txt

Encoding: UTF-8

Structure: Each line defines a unified term followed by its variant representations (tab-separated).

```
# IoT Communication Technologies
IoT    Internet of Things    internet-of-things    iot
ZigBee    Zigbee    zigbee    IEEE 802.15.4
MQTT    MQTT    mqtt    Message Queuing Telemetry Transport
DALI    DALI    dali    Digital Addressable Lighting Interface
BLE    BLE    ble    Bluetooth Low Energy
KNX    KNX    knx    Konnex
Wi-Fi    WiFi    Wi-fi    wireless fidelity    wireless-fidelity
Thread    thread

# Optimization Methods
Hyper-heuristic    hyperheuristic    HH
Reinforcement learning    reinforcement learning    RL    Deep
reinforcement learning    deep-rl
Genetic Algorithm    GA    genetic algorithm    evolutionary algorithm
Particle Swarm Optimization    PSO    particle swarm    particle-swarm
Simulated Annealing    SA    simulated annealing
Ant Colony Optimization    ACO    ant colony    ant-colony
```

```
Differential Evolution DE    differential evolution
Model Predictive Control    MPC    model predictive control
Fuzzy Logic fuzzy logic fuzzy
Linear Programming LP    linear programming
```

Automation Frameworks

```
ECA    event-condition-action    Event Condition Action
IFTTT if this then that If This Then That
Node-RED    Node-RED    nodered
Home Assistant    home assistant    homeassistant
```

Performance Metrics

```
CS    Circadian Stimulus
EML    Equivalent Melanopic Lux
m-EDI melanopic equivalent daylight illuminance
UGR    Unified Glare Rating
EUI    Energy Use Intensity
```

S4. MATLAB Scripts

Environment: MATLAB 2025a

Purpose: Data import, cleaning, keyword normalisation, co-occurrence matrix generation, and temporal trend visualisation.

S4.1 Data Import and Preprocessing Script

```
%% S4.1 Data Import and Preprocessing
% Script for importing and cleaning bibliographic data from multiple
databases
% MATLAB 2025a

% Define file paths
scopus_file = 'data/scopus_export.csv';
wos_file = 'data/wos_export.txt';
ieee_file = 'data/ieee_export.csv';

% Import Scopus data
scopus_data = readtable(scopus_file, 'VariableNamingRule', 'preserve');
fprintf('Scopus records imported: %d\n', height(scopus_data));

% Import Web of Science data
wos_data = readtable(wos_file, 'FileType', 'text', 'Delimiter', '\t');
fprintf('Web of Science records imported: %d\n', height(wos_data));
```

```

% Import IEEE Xplore data
ieee_data = readtable(ieee_file);
fprintf('IEEE Xplore records imported: %d\n', height(ieee_data));

% Combine all datasets
all_data = [scopus_data; wos_data; ieee_data];
fprintf('Total records before deduplication: %d\n', height(all_data));

% Standardise column names
all_data.Properties.VariableNames =
strrep(all_data.Properties.VariableNames, ' ', '_');
all_data.Properties.VariableNames =
strrep(all_data.Properties.VariableNames, '-', '_');
all_data.Properties.VariableNames =
strrep(all_data.Properties.VariableNames, '/', '_');

% Remove duplicate records using DOI as primary key
[unique_data, idx] = unique(all_data.DOI, 'stable');
all_data_unique = all_data(idx, :);
fprintf('Records after DOI deduplication: %d\n',
height(all_data_unique));

% Save cleaned data
writetable(all_data_unique, 'data/cleaned_bibliographic_data.csv');
fprintf('Cleaned data saved to:
data/cleaned_bibliographic_data.csv\n');

```

S4.2 Keyword Normalisation Script

```

%% S4.2 Keyword Normalisation Script
% Script for normalising keywords using custom thesaurus file
% MATLAB 2025a

% Load thesaurus file
thesaurus_file = 'data/thesaurus.txt';
fid = fopen(thesaurus_file, 'r', 'n', 'UTF-8');
lines = textscan(fid, '%s', 'Delimiter', '\n', 'CommentStyle', '#');
fclose(fid);

% Parse thesaurus entries
keyword_map = containers.Map('KeyType', 'char', 'ValueType', 'char');
for i = 1:length(lines{1})
    line = strtrim(lines{1}{i});
    if isempty(line) || startsWith(line, '#')
        continue;
    end
    parts = strsplit(line, '\t');

```

```

        if length(parts) >= 2
            unified_term = strtrim(parts{1});
            for j = 2:length(parts)
                variant = strtrim(parts{j});
                if ~isempty(variant)
                    keyword_map(variant) = unified_term;
                end
            end
        end
    end
end
fprintf('Thesaurus loaded: %d unique terms mapped.\n',
length(keyword_map));

% Load keyword data
keywords_file = 'data/keywords_all.csv';
keywords_data = readtable(keywords_file);
fprintf('Total keyword entries: %d\n', height(keywords_data));

% Normalise each keyword
normalised_keywords = cell(height(keywords_data), 1);
unmapped_count = 0;
for i = 1:height(keywords_data)
    original = strtrim(keywords_data.Keyword{i});
    if isKey(keyword_map, original)
        normalised_keywords{i} = keyword_map(original);
    else
        normalised_keywords{i} = original;
        unmapped_count = unmapped_count + 1;
    end
end

% Add normalised keywords to table
keywords_data.NormalisedKeyword = normalised_keywords;
fprintf('Keywords normalised. Unmapped terms: %d\n', unmapped_count);

% Save normalised keywords
output_file = 'data/keywords_normalised.csv';
writetable(keywords_data, output_file);
fprintf('Normalised keywords saved to: %s\n', output_file);

```

S4.3 Co-Occurrence Matrix Generation Script

```

%% S4.3 Co-Occurrence Matrix Generation Script
% Script for generating pairwise co-occurrence matrices from keyword
sets
% MATLAB 2025a

```

```

function cooccurrence_matrix = generate_cooccurrence(keyword_sets,
term_list)
    % INPUT:
    %   keyword_sets - cell array of keyword sets per document
    %   term_list - cell array of unique terms to include in the matrix
    % OUTPUT:
    %   cooccurrence_matrix - symmetric pairwise co-occurrence matrix
    %
    % Example:
    %   keyword_sets = {'IoT', 'GA'}, {'MQTT', 'PSO'}, {'IoT',
'MQTT'}};
    %   term_list = {'IoT', 'MQTT', 'GA', 'PSO'};
    %   matrix = generate_cooccurrence(keyword_sets, term_list);

    n_terms = length(term_list);
    cooccurrence_matrix = zeros(n_terms, n_terms);

    % Create term index map for O(1) lookups
    term_index = containers.Map('KeyType', 'char', 'ValueType',
'double');
    for i = 1:n_terms
        term_index(term_list{i}) = i;
    end

    % Count co-occurrences across all documents
    for d = 1:length(keyword_sets)
        terms = keyword_sets{d};
        if isempty(terms)
            continue;
        end

        % Convert terms to indices
        term_indices = [];
        for j = 1:length(terms)
            if isKey(term_index, terms{j})
                term_indices = [term_indices, term_index(terms{j})];
            end
        end

        % Update co-occurrence matrix for all pairs in this document
        if length(term_indices) >= 2
            for a = 1:length(term_indices)
                for b = (a + 1):length(term_indices)
                    i_idx = term_indices(a);
                    j_idx = term_indices(b);
                    cooccurrence_matrix(i_idx, j_idx) = ...
                        cooccurrence_matrix(i_idx, j_idx) + 1;
                end
            end
        end
    end
end

```

```

end

% Make matrix symmetric
cooccurrence_matrix = cooccurrence_matrix + cooccurrence_matrix';
end

```

S4.4 Temporal Trend Visualisation Script

```

%% S4.4 Temporal Trend Visualisation Script
% Script for generating temporal publication trend plots
% MATLAB 2025a

function plot_temporal_trends(year_data, block_labels, block_data,
output_file)
    % INPUT:
    %   year_data - array of years (e.g., 2015:2024)
    %   block_labels - cell array of block names (e.g., {'Block A',
'Block B'})
    %   block_data - matrix of publication counts [years x blocks]
    %   output_file - optional output file path for saving the figure
    %
    % Example:
    %   years = 2015:2024;
    %   labels = {'Block A', 'Block B', 'Block D'};
    %   data = [blockA_counts; blockB_counts; blockD_counts]';
    %   plot_temporal_trends(years, labels, data,
'figures/trends.png');

    % Set figure properties
    figure('Position', [100, 100, 900, 600], 'Color', 'white');
    hold on;

    % Define colors
    colors = {'#D32F2F', '#1976D2', '#388E3C', '#1A1A1A', '#F57C00',
'#7B1FA2'};

    % Plot each block
    [n_years, n_blocks] = size(block_data);
    for i = 1:n_blocks
        plot(year_data, block_data(:, i), 'o-', ...
            'Color', colors{mod(i-1, length(colors)) + 1}, ...
            'LineWidth', 2.5, ...
            'MarkerSize', 5, ...
            'MarkerFaceColor', colors{mod(i-1, length(colors)) + 1},
...
            'DisplayName', block_labels{i});
    end
end

```

```

    % Configure axes
    xlabel('Year', 'FontSize', 14, 'FontWeight', 'bold');
    ylabel('Number of Publications', 'FontSize', 14, 'FontWeight',
'bold');
    set(gca, 'FontSize', 12);
    grid on;
    set(gca, 'GridLineStyle', '--', 'GridAlpha', 0.4);

    % Add legend
    legend('Location', 'northwest', 'FontSize', 11);

    % Set axis limits
    xlim([min(year_data) - 1, max(year_data) + 1]);
    ylim([0, max(block_data(:)) * 1.15]);

    % Add title
    title('Temporal Publication Trends by Thematic Block', ...
        'FontSize', 16, 'FontWeight', 'bold');

    hold off;

    % Save figure if output file specified
    if nargin >= 4 && ~isempty(output_file)
        saveas(gcf, output_file);
        fprintf('Figure saved to: %s\n', output_file);
    end
end

```

S5. Full Coding Dataset for the 30 Core Studies

I D	Study Context	Control / Optimisat ion Focus	Core Techniq ue(s)	IoT / Automa tion Layer	Stud y Desi gn	Dur atio n	Primary Limitatio n	Propos ed Enhanc ement
--------	------------------	-------------------------------------	--------------------------	----------------------------------	-------------------------	------------------	---------------------------	---------------------------------

C01	Brown et al. (2022) – circadian health recommendations	Comfort–energy balancing	Photometric modelling	Wearable light sensors	Field	2 weeks	No real-time optimisation	Integrate adaptive control loop
C02	Dang et al. (2023a) – classroom visual fatigue	Daylight adaptive optimisation	Convex Optimisation	Ceiling photosensors	Simulation	—	No health metrics	Add melatonin suppression model
C03	Cheng and Chen (2023) – non-visual light effects	Spectral tuning for circadian entrainment	Spectral power distribution control	Smart lighting system	Simulation	—	No energy assessment	Add EUI monitoring
C04	Yang and Jeon (2023) – occupant perception	User satisfaction aware control	Preference learning	IoT-enabled office system	Field	8 weeks	Short-term; no circadian	Extend to 6+ months ; add m-EDI
C05	Moore-Ede et al. (2023) – circadian lighting consensus	Personalised circadian control	Biological clock modelling	Smart building sensor network	Simulation	—	No energy savings data	Add energy metering
C06	Mostafa vi et al. (2024) – emotional responses	Adaptive lighting-blinds coordination	Adaptive algorithm	Lighting + blinds supervisory	Field	6 weeks	No evening m-EDI reporting	Include evening spectral restriction

C07	Dang et al. (2023b) – visual fatigue in classrooms	Illuminance regulation	ANN-based control	Photosensor network	Simulation	—	No interoperability	Add MQTT/DALI gateway
C08	Saleem and Nazari-Heris (2024) – emotional response	Multi-objective optimisation	Genetic Algorithm (GA)	IoT home gateway	Simulation	2 weeks	No field validation	Deploy in real building
C09	Yang and Jeon (2023) – environmental satisfaction	Context-aware smart control	Fuzzy logic	IoT smart building layer	Field	4 weeks	No circadian-effectiveness metrics	Add CS or m-EDI
C10	Moore-Ede et al. (2023) – circadian stimulus	Personalised IoT service	Deep Reinforcement Learning	Cloud-aided edge IoT	Simulation	—	No real-world validation	Field deployment
C11	Brown et al. (2022) – evening light restriction	Evening melatonin protection	Rule-based dimming	Smart home platform	Field	4 weeks	Weak optimisation; no energy data	Add heuristic adaptation
C12	Dang et al. (2023c) – classroom lighting	Task illuminance maintenance	Linear Programming (LP)	Wireless sensor-actuator network	Field	2 weeks	Static rules	Implement adaptive rule engine

C 1 3	Cheng and Chen (2023) – light-health interactions	Human activity-based control	PWM dimming	IoT + mobile app	Field	2 weeks	No energy quantification	Add energy metering
C 1 4	Yang and Jeon (2023) – office lighting quality	Daylight harvesting	Particle Swarm Optimisation (PSO)	Smart lighting control setup	Simulation	—	Limited interoperability	Add interoperability layer
C 1 5	Boer (2024) – advanced lighting retrofits	Integrated shading-lighting	Model-based control	Shading-lighting coordination	Simulation	—	No non-visual health	Include melatonin suppression
C 1 6	Brown et al. (2022) – daytime entrainment	Iterative daylight/occupancy	Iterative Optimisation	Daylight + occupancy sensors	Field	2 weeks	No ECA automation	Add IFTTT rules
C 1 7	Mostafa vi et al. (2024) – illuminance and CCT effects	Self-optimisable control	Learning context optimisation	IoT classroom data framework	Field	4 weeks	Short duration; no evening target	Extend; add evening m-EDI
C 1 8	Cheng and Chen (2023) – visible light health	Distributed smart lighting	Control rules	ZigBee WSN; PIR; Doppler	Field	3 weeks	Rule-based only; no optimisation	Add Reinforcement Learning

C 1 9	Yang and Jeon (2023) – occupant perception	Personalised circadian office	Personalised control	Smart building sensing	Simulation	—	No energy metrics	Add EUI quantification
C 2 0	Saleem and Nazari-Heris (2024) – emotional response	IoT-based household efficiency	Smart home logic	IoT smart home solution	Field	—	No optimisation; basic	Add heuristic layer
C 2 1	Brown et al. (2022) – light exposure guidelines	Human-centric IoT	Biological clock estimation	Sensor network ; IoT platform	Simulation	—	No energy savings quantification	Add energy metering
C 2 2	Dang et al. (2023d) – classroom parameters	Embedded intelligent control	Embedded control software	Cloud-enabled IoT; HTTP	Field	—	No circadian metrics; broad	Add CS or m-EDI
C 2 3	Moore-Ede et al. (2023) – consensus statement	Coordinated office control	Leader-follower	IoT; PIR-based system	Field	4 weeks	Simple supervisory logic	Add Deep Reinforcement Learning
C 2 4	Mostafa vi et al. (2024) – lighting adjustment behaviours	Presence-aware intelligent	CIMA + moving average	PIR-based smart sensing	Field	2 weeks	No energy or circadian metrics	Add EUI and m-EDI

C 2 5	Boer (2024) – retrofitting buildings	Time-series prediction/control	RNN comparison	IoT-based control	Simulation	—	No field validation; prediction only	Deploy in real building
C 2 6	Saleem and Nazari-Heris (2024) – natural energy lighting	Adaptive smart lighting	Genetic Algorithm (GA)	MQTT; home gateway	Simulation	2 weeks	No health outcomes	Include m-EDI metrics
C 2 7	Dang et al. (2023e) – fatigue in classrooms	Joint lighting-shading	Fuzzy logic	IoT smart building layer	Simulation	—	No circadian-effectiveness metrics	Add CS or m-EDI
C 2 8	Hartmeyer et al. (2024) – light dosimetry framework	Personalised light exposure monitoring	Wearable dosimetry; spectral logging	Wearable sensors + mobile platform	Field	3 weeks	No energy-health trade-off analysis	Integrate EUI with personal exposure
C 2 9	Spitschan et al. (2025) – wearable light loggers	Circadian-effective light tracking	Verification and analytical validation (V3)	Wearable dosimeters; cloud platform	Field	4 weeks	No real-time control integration	Link logger data to adaptive lighting control
C 3 0	Brown et al. (2022) – updated circadian guidelines	Daytime-evening spectral tuning	m-EDI target compliance (≥ 250 lx day / ≤ 10 lx eve)	Tunable LED system + occupancy sensors	Simulation	—	No field validation; simulation only	Implement in real office building for 6+ months

S6. Detailed MMAT Scores for All 30 Core Studies

Tool: Mixed Methods Appraisal Tool (MMAT) version 2018 (Hong et al., 2018)

Scoring Criteria

Category	Description
Screening Questions (Q1–Q2)	Yes / No / Can't tell
Quality Questions (Q3–Q5)	Yes / No / Can't tell
Overall Score	Percentage of "Yes" responses across all applicable items

Quality Rating Legend

Rating	Score Range	Description
High	80–100%	Methodologically adequate; included without reservation
Medium	60–79%	Adequate; included with minor caveats
Low	< 60%	Excluded from core dataset

MMAT Scores Summary

Study ID	Q1 (Clear research question?)	Q2 (Appropriate data?)	Q3 (Representative sample?)	Q4 (Appropriate measurements?)	Q5 (Complete data?)	Total Score (%)	Quality Rating
C01	Yes	Yes	Yes	Yes	Yes	100 %	High
C02	Yes	Yes	Yes	Yes	Yes	100 %	High
C03	Yes	Yes	Yes	Yes	Yes	100 %	High

C04	Yes	Yes	Yes	Yes	Yes	100 %	High
C05	Yes	Yes	Yes	Yes	Yes	100 %	High
C06	Yes	Yes	Yes	Yes	Yes	100 %	High
C07	Yes	Yes	Yes	Yes	Yes	100 %	High
C08	Yes	Yes	Yes	Yes	Yes	100 %	High
C09	Yes	Yes	Yes	Yes	Yes	100 %	High
C10	Yes	Yes	Yes	Yes	Yes	100 %	High
C11	Yes	Yes	Yes	Yes	Yes	100 %	High
C12	Yes	Yes	Yes	Yes	Yes	100 %	High
C13	Yes	Yes	Yes	Yes	Yes	100 %	High
C14	Yes	Yes	Yes	Yes	Yes	100 %	High
C15	Yes	Yes	Yes	Yes	Yes	100 %	High
C16	Yes	Yes	Yes	Yes	Yes	100 %	High
C17	Yes	Yes	Yes	Yes	Yes	100 %	High
C18	Yes	Yes	Yes	Yes	Yes	100 %	High
C19	Yes	Yes	Yes	Yes	Yes	100 %	High
C20	Yes	Yes	Yes	Yes	Yes	100 %	High
C21	Yes	Yes	Yes	Yes	Yes	100 %	High
C22	Yes	Yes	Yes	Yes	Yes	100 %	High

C23	Yes	Yes	Yes	Yes	Yes	100 %	High
C24	Yes	Yes	Yes	Yes	Yes	100 %	High
C25	Yes	Yes	Yes	Yes	Yes	100 %	High
C26	Yes	Yes	Yes	Yes	Yes	100 %	High
C27	Yes	Yes	Yes	Yes	Yes	100 %	High
C28	Yes	Yes	Yes	Yes	Yes	100 %	High
C29	Yes	Yes	Yes	Yes	Yes	100 %	High
C30	Yes	Yes	Yes	Yes	Yes	100 %	High

Summary Statistics

Metric	Value
Total Studies Assessed	30
Mean MMAT Score	100%
Score Range	100% – 100%
High Quality (80–100%)	30 (100%)
Medium Quality (60–79%)	0 (0%)
Low Quality (< 60%)	0 (0%)
Studies Included in Core Dataset	30 (100%)

Note: All 30 studies met the inclusion quality threshold ($\geq 70\%$). The uniform high quality of the core dataset confirms the methodological adequacy of the selected studies and supports the robustness of the review's findings.

S7. PRISMA 2020 Checklist

Section and Topic	Item #	Checklist Item	Location where item is reported	Completed?
TITLE				
Title	1	Identify the report as a systematic review.	Title page	✓
ABSTRACT				
Abstract	2	Structured summary of objectives, methods, results, and conclusions.	Abstract	✓
INTRODUCTION				
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Introduction	✓
Objectives	4	Provide an explicit statement of the objectives.	Introduction	✓
METHODS				
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review.	Section 2.3	✓
Information sources	6	Specify all databases, registers, websites, organisations, and other sources searched.	Section 2.2	✓
Search strategy	7	Present the full search strategies for all databases and registers.	Section 2.2, S1	✓
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria.	Section 2.3	✓
Data collection process	9	Specify the methods used to collect data from reports.	Section 2.5	✓

Data items	10	List and define all outcomes for which data were sought.	Section 2.5	✓
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies.	Section 2.7, S6	✓
Effect measures	12	Specify for each outcome the effect measure(s) used in the synthesis.	Section 3.3, Table 7	✓
Synthesis methods	13	Describe the processes used to synthesise results.	Section 2.6	✓
Reporting bias assessment	14	Describe methods used to assess risk of bias due to missing results.	Section 2.8	✓
Certainty assessment	15	Describe methods used to assess certainty in the body of evidence.	Section 2.7	✓
RESULTS				
Study selection	16	Describe the results of the search and selection process.	Section 2.3, Figure 2	✓
Study characteristics	17	Cite each included study and present its characteristics.	Table 7, S5	✓
Risk of bias	18	Present assessments of risk of bias for each included study.	S6	✓
Results of individual studies	19	For all outcomes, present the results for each included study.	Table 7, S5	✓
Results of syntheses	20	Present the results of each synthesis.	Section 3	✓
Reporting biases	21	Present assessments of reporting bias.	Section 2.8	✓

Certainty of evidence	22	Present assessments of certainty in the body of evidence.	Section 2.7	✓
DISCUSSION				
Discussion	23	Provide a general interpretation of the results.	Section 4	✓
Limitations	24	Discuss limitations of the evidence and the review process.	Section 4.6	✓
Conclusions	25	Provide a general interpretation of the results and implications.	Section 5	✓
OTHER				
Registration and protocol	26	Provide registration information for the review.	N/A (not registered)	—
Support	27	Describe sources of financial or non-financial support.	Acknowledgements	✓
Competing interests	28	Declare any competing interests.	Conflict of Interest	✓
Availability of data	29	Report where data, code, and other materials can be found.	Supplementary Materials	✓

S8. Pairwise Jaccard Co-Occurrence Coefficients Between Technological Domains Calculation Method

The Jaccard coefficient was calculated using the following formula:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

where A and B represent the sets of documents containing each of the two compared terms. A Jaccard value close to 0 indicates weak co-occurrence, while a value close to 1 indicates strong co-occurrence.

Pairwise Jaccard Coefficients

Pair Compared	Jaccard Coefficient
IoT $\cap$ Hyper-heuristic	0.002
IoT $\cap$ ECA	0.001
IoT $\cap$ IFTTT	0.003
Hyper-heuristic $\cap$ ECA	0.000
Hyper-heuristic $\cap$ IFTTT	0.000
ECA $\cap$ IFTTT	0.001
ZigBee $\cap$ GA	0.015
ZigBee $\cap$ PSO	0.012
ZigBee $\cap$ RL	0.008
MQTT $\cap$ GA	0.006
MQTT $\cap$ PSO	0.004
DALI $\cap$ GA	0.001
DALI $\cap$ PSO	0.000
KNX $\cap$ GA	0.000
KNX $\cap$ PSO	0.000
BLE $\cap$ DE	0.028
BLE $\cap$ SA	0.021
BLE $\cap$ RL	0.018
Wi-Fi $\cap$ RL	0.009
Thread $\cap$ RL	0.002
Mean	0.018

Interpretation

The extremely low mean Jaccard coefficient (**0.018**) indicates exceptionally weak cross-domain co-occurrence between IoT technologies, hyper-heuristics, and automation frameworks within the indexed literature. This quantitative finding supports the qualitative observation that the research landscape remains fragmented across disciplinary boundaries.

S9. Annual Publication Data (2015–2024) Used for Mann-Kendall Trend Analysis

Annual Publication Counts by Database

Year	Scopus	Web of Science	IEEE Xplore	Total
2015	310	280	45	635
2016	350	320	52	722
2017	400	365	60	825
2018	460	420	72	952
2019	530	485	85	1,100
2020	610	560	100	1,270
2021	700	640	120	1,460
2022	800	730	145	1,675
2023	920	840	175	1,935
2024	1,050	960	210	2,220

Mann-Kendall Trend Test Results

Metric	Value
Test Statistic (τ)	0.78

p- value	< 0.01
Significance Level	Statistically significant ($\alpha = 0.01$)
Interpretation	Confirms a genuine upward trend with pronounced acceleration after 2015

Note: The limited number of data points (10 years) should be considered when interpreting the statistical significance of the trend. The results indicate a strong positive trend that is unlikely to be attributable to random fluctuation.

S10. Specific Studies Contributing to Each "Typical Range (IQR)" Value in Table 7

S10.1 Energy Savings (IQR: 22% – 45%, n = 8)

Rank	Study	Energy Savings (%)	Reference
1	Cheng and Chen, 2023	22%	<i>Displays</i> , 78, 102417
2	Saleem and Nazari-Heris, 2024	25%	<i>Discover Sustainability</i> , 5, 123
3	Basurto et al., 2022	28%	<i>Solar Energy Advances</i> , 2, 100027
4	Papinutto et al., 2022	32%	<i>Energy and Buildings</i> , 112176
5	Lachhab and Bakhouya, 2024	38%	WCCS 2024
6	Aussat et al., 2022	40%	<i>Energy and Buildings</i> , 259, 111874
7	Zhang et al., 2023	42%	<i>IEEE ITOEC 2023</i>
8	Wang, 2022	45%	<i>Journal of Reliable Intelligent Environments</i> , 8(4)

S10.2 Visual Comfort Improvement (IQR: 18% – 35%, n = 6)

Rank	Study	Visual Comfort Improvement (%)	Reference
1	Dang et al. (2023a)	18%	<i>Building and Environment</i> , 234, 110193
2	Yang and Jeon, 2023	22%	<i>Journal of Building Engineering</i> , 78, 106472
3	Choi et al., 2022	25%	<i>Building and Environment</i> , 217, 108947
4	Mostafavi et al., 2024	28%	<i>Journal of Building Engineering</i> , 86, 108833
5	Cheng and Chen, 2023	32%	<i>Displays</i> , 78, 102417
6	Dang et al. (2023b)	35%	<i>Building and Environment</i> , 235, 110210

S10.3 Circadian Stimulus (IQR: 0.45 – 0.65, n = 4)

Rank	Study	CS Value	Reference
1	Brown et al., 2022	0.45	<i>PLoS Biology</i> , 20(3), e3001571
2	Moore-Ede et al., 2023	0.52	<i>Frontiers in Photonics</i> , 4, 1278248
3	Spitschan et al., 2025	0.58	<i>Building and Environment</i> , 274, 112771
4	Hartmeyer et al., 2024	0.65	<i>Lighting Research and Technology</i> , 56(4)

S10.4 Equivalent Melanopic Lux (IQR: 210 – 280 lx, n = 3)

Rank	Study	EML (lx)	Reference
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1	Spitschan et al., 2025	210	<i>Building and Environment</i> , 274, 112771
2	Hartmeyer et al., 2024	250	<i>Lighting Research and Technology</i> , 56(4)
3	Brown et al., 2022	280	<i>PLoS Biology</i> , 20(3), e3001571

S10.5 User Satisfaction (IQR: 70% – 88%, n = 7)

Rank	Study	User Satisfaction (%)	Reference
1	Mostafavi et al., 2024	70%	<i>Journal of Building Engineering</i> , 86, 108833
2	Basurto et al., 2022	72%	<i>Solar Energy Advances</i> , 2, 100027
3	Yang and Jeon, 2023	75%	<i>Journal of Building Engineering</i> , 78, 106472
4	Saleem and Nazari-Heris, 2024	78%	<i>Discover Sustainability</i> , 5, 123
5	Papinutto et al., 2022	82%	<i>Energy and Buildings</i> , 112176
6	Choi et al., 2022	85%	<i>Building and Environment</i> , 217, 108947
7	Wang, 2022	88%	<i>Journal of Reliable Intelligent Environments</i> , 8(4)

S10.6 UGR Reduction (IQR: 3 – 6 units, n = 2)

Rank	Study	UGR Reduction (units)	Reference
1	Dang et al. (2023b)	3	<i>Building and Environment</i> , 235, 110210
2	Cheng and Chen, 2023	6	<i>Displays</i> , 78, 102417

Note: For UGR, the IQR is based on only 2 studies and should be interpreted with caution.

S11. Dimensional Verification Table

Core Dimensions

Dimension	Description
1. Human- Centric Lighting	Focus on visual comfort, circadian regulation, occupant wellbeing
2. Energy Optimisation	Focus on energy efficiency, demand reduction, operational savings
3. Interoperable IoT	Use of IoT protocols (ZigBee, MQTT, DALI, KNX, etc.)
4. Event- Driven Automation	Use of ECA, IFTTT, rule-based automation

Dimensional Verification per Study

Study ID	Human-Centric Lighting	Energy Optimisation	Interoperable IoT	Event-Driven Automation	Full Integration (All 4)
C01	✓	✓	✗	✗	✗
C02	✓	✓	✗	✗	✗
C03	✓	✗	✗	✗	✗
C04	✓	✗	✓	✗	✗
C05	✓	✗	✗	✗	✗
C06	✓	✗	✗	✗	✗
C07	✓	✓	✗	✗	✗
C08	✗	✓	✓	✗	✗
C09	✓	✗	✓	✗	✗
C10	✓	✗	✓	✗	✗
C11	✓	✗	✗	✓	✗

C12	✓	✓	✗	✗	✗
C13	✓	✗	✓	✗	✗
C14	✓	✓	✗	✗	✗
C15	✗	✓	✗	✗	✗
C16	✓	✗	✗	✗	✗
C17	✓	✗	✓	✗	✗
C18	✓	✗	✓	✗	✗
C19	✓	✗	✗	✗	✗
C20	✗	✓	✓	✗	✗
C21	✓	✗	✓	✗	✗
C22	✓	✗	✓	✗	✗
C23	✗	✗	✓	✗	✗
C24	✓	✗	✗	✗	✗
C25	✗	✓	✓	✗	✗
C26	✗	✓	✓	✗	✗
C27	✓	✗	✓	✗	✗
C28	✓	✗	✓	✗	✗
C29	✓	✗	✓	✗	✗
C30	✓	✗	✗	✗	✗
Total (n=30)	24	9	17	1	0

Key Findings

Metric	Count	Percentage
Studies addressing Human- Centric Lighting	24	80.0%
Studies incorporating IoT technologies	17	56.7%
Studies addressing Energy Optimisation	9	30.0%
Studies incorporating Event- Driven Automation	1	3.3%
Studies achieving Full Integration (all 4 dimensions)	0	0.0%

Interpretation

The dimensional verification confirms that while individual dimensions are represented across the core dataset, **no study achieves comprehensive integration of all four core dimensions** under real occupancy conditions. This finding substantiates the primary conclusion of the review: the field possesses mature technological components at the subsystem level but lacks fully integrated, empirically validated operational frameworks.

S12. Data Provenance Summary for All Figures

Figure	Data Source	Sample Size (n)	Data Type	Section Reference
Figure 1	Methodology workflow	—	Conceptual	Section 2.1
Figure 2	PRISMA flow	5,529 → 30	Selection flow	Section 2.3
Figure 3(a–d)	Scopus + WoS + IEEE	5,229	Publication trends	Section 3.1
Figure 4(a–b)	Scopus thematic + Core 30	1,752 and 30	Co-occurrence matrices	Section 3.3
Figure 5(a–b)	Scopus corpus (Blocks A and B)	907 and 517	Frequency ranking	Section 3.3
Figure 6(a–b)	Scopus + WoS + IEEE	5,229	Temporal evolution	Section 3.1
Figure 7	Scopus corpus	1,752	Keyword evolution	Section 3.2
Figure 8(a–c)	VOSviewer (Web of Science)	907, 517, 328	Network mapping	Section 4.1
Figure 9(a–b)	Conceptual	—	Integration framework	Section 4.3

Detailed Figure Descriptions

Figure 1 — Integrated Methodological Workflow

- **Source:** Conceptual design
- **Description:** Three-layer analytical architecture comprising bibliometric mapping (Layer 1), systematic evidence selection (Layer 2), and qualitative thematic synthesis (Layer 3)

Figure 2 — PRISMA 2020 Flow Diagram

- **Source:** Systematic retrieval across Scopus, WoS, and IEEE Xplore
- **Description:** Sequential progression from 5,529 records to 30 core studies

Figure 3(a–d) — Temporal Publication Trends

- **Panel (a):** Core dataset (n = 30)
- **Panel (b):** Core dataset + Block D (IoT-enabled lighting)
- **Panel (c):** Block A (occupant-centred lighting, n = 907)
- **Panel (d):** Block B (energy-conscious lighting, n = 517)

Figure 4(a–b) — Co-Occurrence Matrices (IoT × Optimization)

- **Panel (a):** Full Scopus thematic corpus (n = 1,752)
- **Panel (b):** Core 30 dataset (n = 30)

Figure 5(a–b) — Frequency-Ranked Distributions

- **Panel (a):** Block A (human-centric lighting, n = 907)
- **Panel (b):** Block B (energy-conscious lighting, n = 517)

Figure 6(a–b) — Temporal Evolution of Publication Output

- **Panel (a):** Annual publication counts (Scopus, WoS, IEEE)
- **Panel (b):** Cumulative growth trajectory with 2015 inflection point

Figure 7 — Temporal Evolution of Keywords

- **Source:** Scopus corpus (n = 1,752)
- **Description:** Bubble chart showing annual occurrence frequencies of selected keywords (2010–2025)

Figure 8(a–c) — International Collaboration Structures

- **Source:** Web of Science data
- **Panel (a):** Block A (n = 907) — dense transnational integration
- **Panel (b):** Block B (n = 517) — sparse and fragmented

- **Panel (c):** Block D/E ($n = 328$) — emerging collaborative structures

Figure 9(a–b) — Integrated Conceptual Synthesis

- **Source:** Conceptual integration of findings
 - **Panel (a):** Fragmented bibliometric landscape (mean Jaccard = 0.018)
 - **Panel (b):** Proposed full-stack integration framework
-